World Happiness

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Introduction:

- Happiness reflects quality over life
- Not just income but also factors like health, social support, freedom,etc.
- Government uses happiness data for policy decisions



Research Questions:

01 Which factors most influence happiness?

02 How has happiness changed over time

03 Are there geographic patterns in happiness?

04 Can we predict happiness using machine learning?

Data & Methods:

Dataset:
World Happiness Report(2015-2019)
Multiple countries across the world

Variables used:

- GDP per capita
- Social Support
- Health
- Freedom
- Generosity

Methods:
Data cleaning(merge + handle missing values)
Visualization(heatmap,line chart, scatter plot)
Machine Learning(Random Forest model)

```
[14]: print(df.columns)
      print(df.head)
```

```
Index(['Happiness Score', 'GDP per capita', 'Social support',
       'Healthy life expectancy', 'Freedom', 'Generosity',
       'Perceptions of corruption', 'Year'],
      dtype='object')
<bound method NDFrame.head of      Happiness Score  GDP per capita  Social support  Healthy life expectancy  \
0              7.587         1.39651        1.211026                  0.94143
1              7.561         1.30232        1.211026                  0.94784
2              7.527         1.32548        1.211026                  0.87464
3              7.522         1.45900        1.211026                  0.88521
4              7.427         1.32629        1.211026                  0.90563
..               ...             ...             ...                      ...
777            3.334         0.35900        0.711000                  0.61400
778            3.231         0.47600        0.885000                  0.49900
779            3.203         0.35000        0.517000                  0.36100
780            3.083         0.02600        0.000000                  0.10500
781            2.853         0.30600        0.575000                  0.29500

      Freedom  Generosity  Perceptions of corruption  Year
0    0.665570     0.29678                   0.111299  2015
1    0.628770     0.43630                   0.111299  2015
2    0.649380     0.34139                   0.111299  2015
3    0.669730     0.34699                   0.111299  2015
4    0.632970     0.45811                   0.111299  2015
..        ...         ...                        ...   ...
777  0.402828     0.21700                   0.411000  2019
778  0.402828     0.27600                   0.147000  2019
779  0.402828     0.15800                   0.025000  2019
780  0.402828     0.23500                   0.035000  2019
781  0.402828     0.20200                   0.091000  2019

[782 rows x 8 columns]>
```

Factors Influencing Happiness:

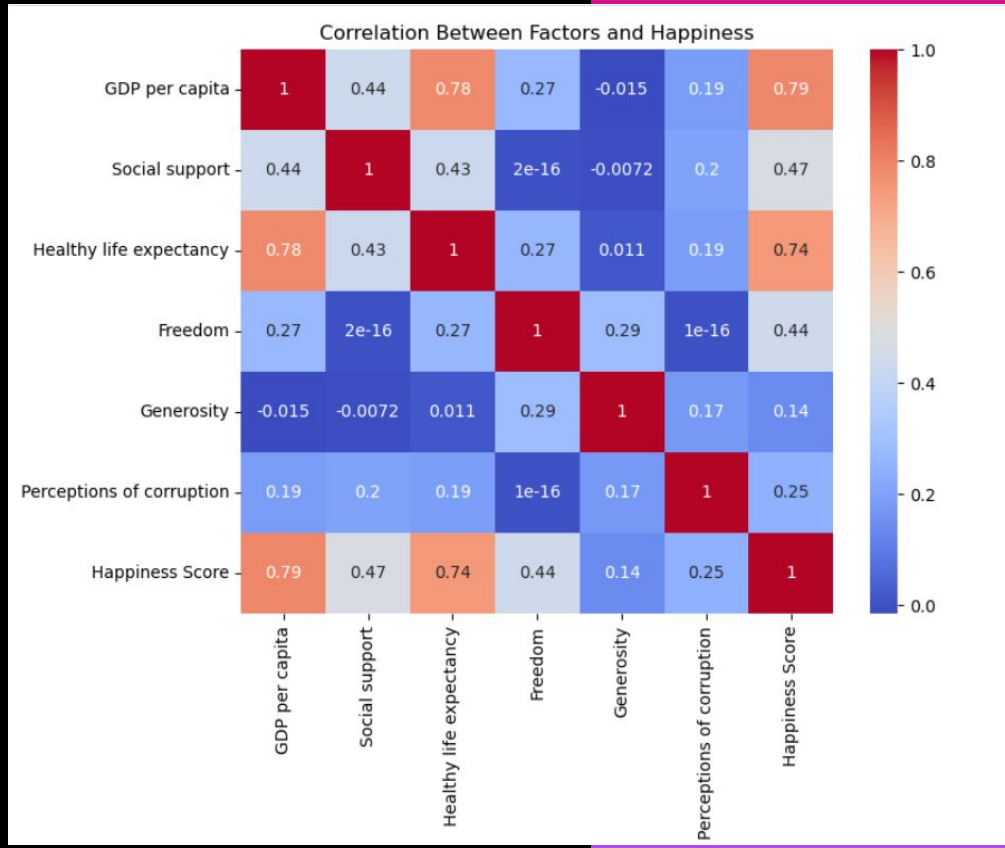
Strong positive relationships:

- GDP per capita
- Social support
- Healthy life expectancy

Weak relationships:

- Generosity
- Corruption

"Darker red = stronger positive relationship"



MACHINE LEARNING (FEATURE IMPORTANCE):

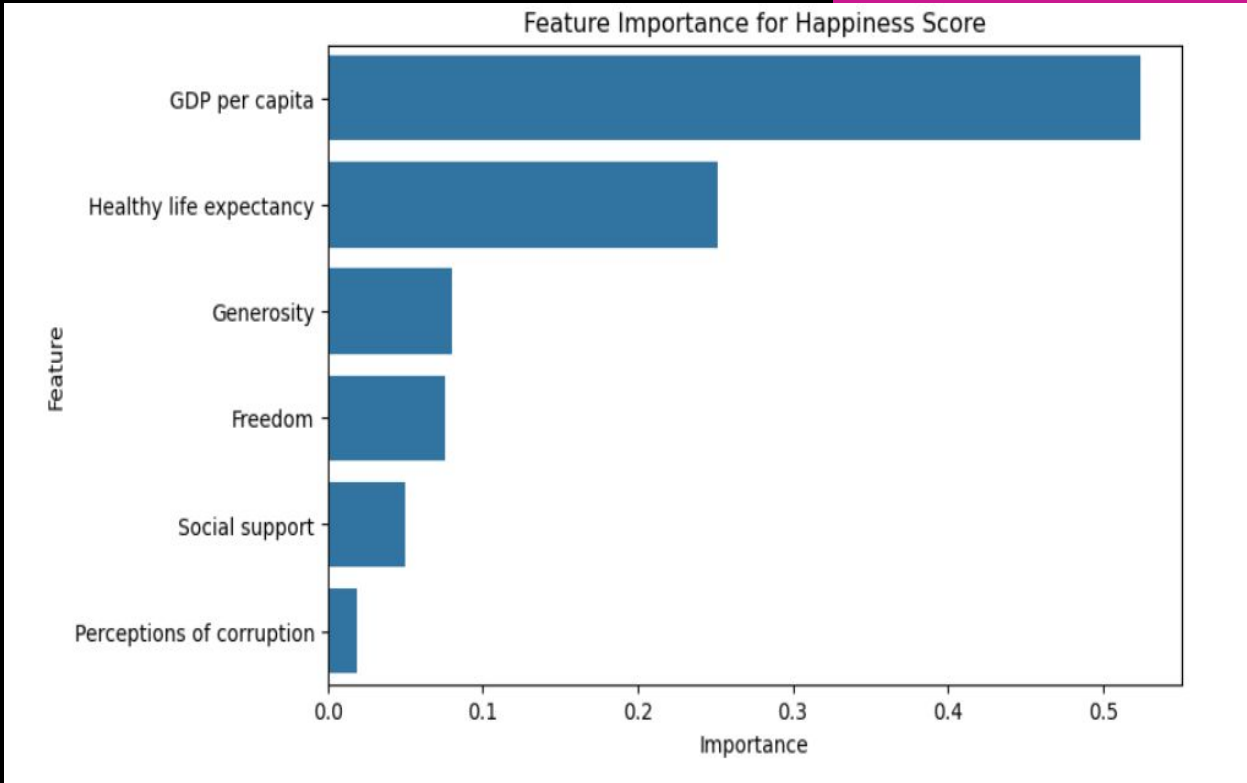
- Model used: Random Forest
- Measures importance of each factor

Key results:

GDP per capita → most important

Social support → second

Health → also important

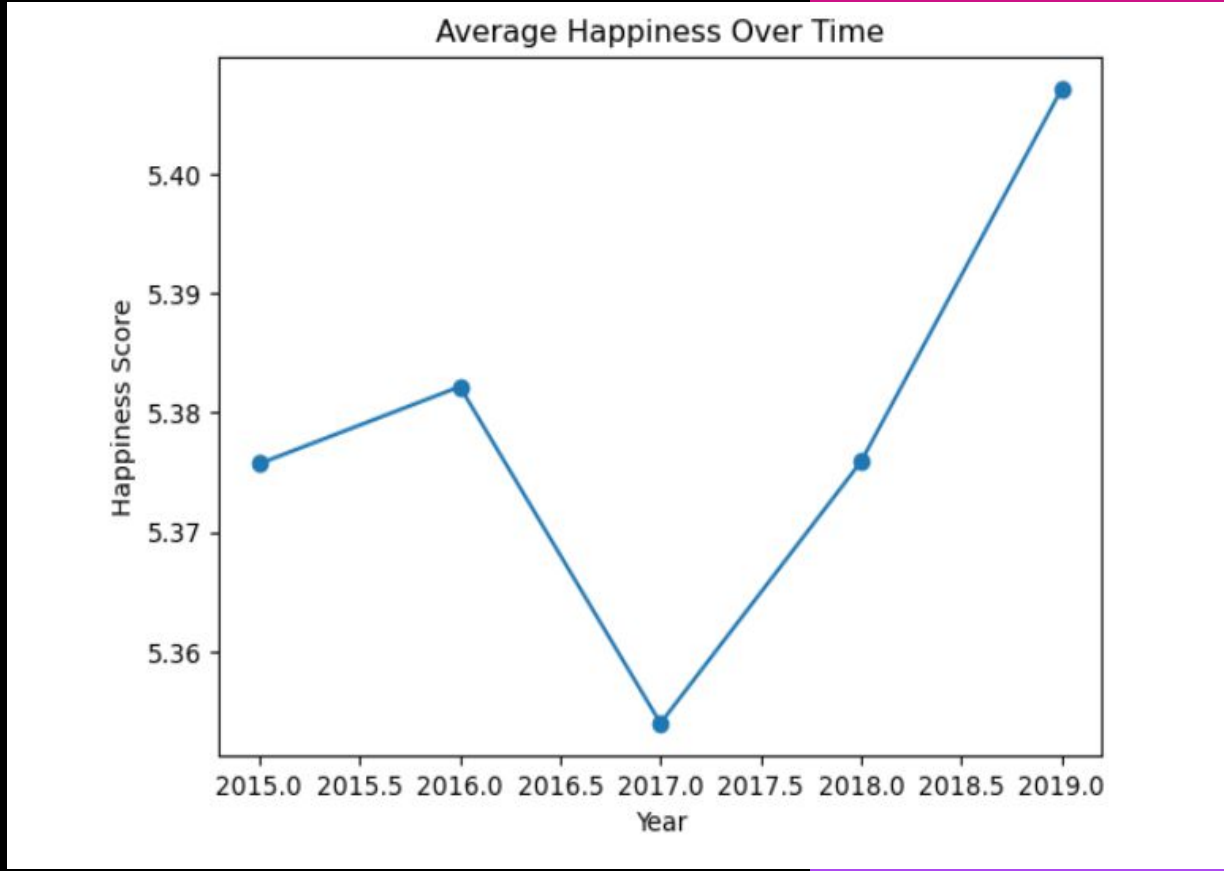


Happiness over time:

Average happiness by year
Data from 2015–2019

Observation:Observation:

- Slight fluctuations
- No major increase or decrease

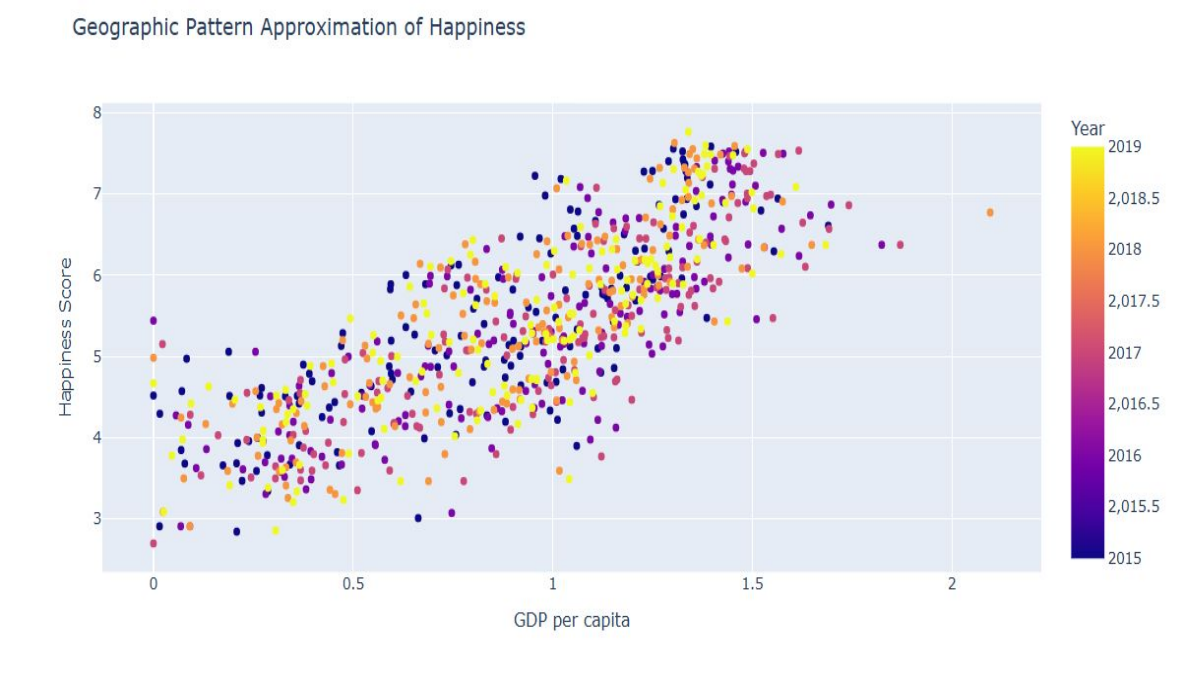


GEOGRAPHIC / GLOBAL PATTERNS:

- Each point = a country
- Higher GDP → higher happiness

Observation:

- Clear upward trend
- Clusters of high-income countries



Model Performance:

- Model predicts happiness using features
- Mean Squared Error (MSE) used to evaluate

```
Mean Squared Error: 0.277760384797527
```

Interpretation:

- Model performs reasonably well
- Shows happiness is predictable

Limitations and Conclusion:

Limitations:

- Only 5 years of data → limited trend analysis
- Missing values filled → may reduce accuracy
- Happiness is subjective → cultural differences

ext Steps:

- Use more years of data
- Include more variables (education, politics)
- Use more advanced models

Conclusion:

Happiness is influenced by measurable factors

Large differences exist across countries

Data helps explain global well-being

